LIKHITA YERRA

AI Engineer & Researcher

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PROFILE

AI researcher and engineer based in Paris (Master's in AI & Data Science, aivancity; first in class PGE4 & PGE5), seeking a PhD in machine learning. My work spans LLM reliability and calibration, uncertainty quantification, multi-objective and sustainability-aware model evaluation, representation learning, digital twins, and multimodal / signal ML. I publish and present in English, build end-to-end research and production systems in Python / PyTorch, and have industry experience shipping LLM and computer-vision products.

RESEARCH INTERESTS

Uncertainty quantification and calibration of large language models; temporal knowledge decay and staleness-awareness; multi-objective LLM evaluation (quality, latency, energy, CO_2, cost); trustworthy and compliance-aware RAG; representation learning and transfer to small datasets; digital twins and adaptive control for cyber-physical systems.

EDUCATION

Master's in AI and Data Science, aivancity School for Technology, Business & Society 09/2023 - 2026 | Paris

- GPA **16.39/20 (2024/25, first in class, Mention Très Bien)**; **17.11/20 (2025/26, first in class, Mention Très Bien)**.
- **PFE (Master's Thesis)**, defended 09/2026, supervised by Antoun Yaacoub: *Reference-Angle Optimization for Identity-Preserving Face Generation*. Learned identity-optimal reference-angle predictor (ArcFace + MobileNetV2); top-2 accuracy of 85% under leave-one-subject-out (LOSO) evaluation.
- **Relevant coursework**: Mathematical Foundations of Deep Learning; Time-Series Analysis; Explainable AI; Fair Machine Learning; Robustness; Data Anonymization; Generative AI; Agentic AI and AI Agents; Green AI.

B.Tech (Hons) in Data Science & Data Engineering, Lovely Professional University (credit transfer into aivancity) 08/2021 - 07/2023 | India / Paris

PUBLICATIONS & PRESENTATIONS

Do Language Models Know When Their Knowledge Expires? Measuring Calibration Decay Across the Knowledge Horizon.
Submitted to Women in NeurIPS 2026. Introduces *calibration aging curves* (accuracy and confidence on dated facts, stratified by fact age relative to the training cutoff). On 174 Wikidata head-of-government facts and three open-weight models (Qwen2.5-1.5B/3B-Instruct, SmolLM2-1.7B-Instruct), accuracy on post-cutoff facts collapses (e.g. 16.9% to 2.4%) while verbalized confidence stays pinned at ~90-95%: a staleness-awareness gap.

CNN-Adaptive Sliding Mode Control for Autonomous Differential-Drive Robots Under Environmental Uncertainty.
IEEE DELCON 2026 (presenting; BITS Pilani; IEEE Xplore). A lightweight CNN classifies the environment from a 64x64 occupancy map and selects an SMC parameter preset; Lyapunov analysis establishes uniform ultimate boundedness. Across five simulated uncertainty scenarios (vs. fixed-gain, fuzzy, oracle and RL baselines): 35.7% less angular-velocity chattering under sensor noise; 95.1% classification accuracy on realistic occupancy maps.

Visualizing Adaptive Control: A 3D Digital Twin Framework for CNN-Supervised SMC in Mobile Robots.
EIRT 2026 (presenting; IEEE). React/Three.js hospital-corridor digital twin over FastAPI/WebSocket for live Classical / CNN-adaptive / PPO comparison, with replay and CSV export.

Compar'IA: An Extensible Sustainability-Aware Dashboard for Multi-Objective Evaluation of Large Language Models.
Green-PR Workshop, ICPR 2026, Lyon, France (presented). Benchmarks LLaMA, Mistral, GPT and DeepSeek on 30 prompts across quality, latency, energy, CO_2 and cost.

Building Trustworthy AI Pipelines: A Compliance-Aware Architecture Using Retrieval-Augmented Generation.
Chapter accepted, forthcoming Elsevier book *Trustworthy LLMs: Principles and Challenges*, March 2027.

Semantic State Abstraction Interfaces for LLM-Augmented Portfolio Decisions.
arXiv. LLM-based semantic state abstraction from financial news for PPO/SAC portfolio agents; multi-axis news decomposition and RL diagnostics.

AI-Powered Chess Insights: Live Move Detection and Narration Using Computer Vision and Large Language Models.
IEEE Cognitive Nexus Magazine, Vol. 1, Issue 1, 2025. YOLOv8 + python-chess + GPT-4 pipeline for chess move detection and voiced commentary.

PROFESSIONAL EXPERIENCE

IAware/amzilai - AI Engineer (Intern, then Apprentice on the Technical Lead track) 05/2025 - Present | Paris

- Delivered end-to-end AI solutions across 5 domains: photography, marketing, aesthetics, real estate and service automation.
- Own the full loop: problem framing, model/API choice, shipping into client workflows, and iterating from production failures.
- Started as an intern (05/2025 - 09/2025) on end-to-end development of industrial AI projects across diverse domains.

AIRecruiter (Rovr) - Lead Data Scientist Intern 07/2024 - 09/2024 | Zurich

- First technical hire; built RAG-LLM pipelines from scratch with the founder and ex-Google engineers.

AI CLINIC (AIVANCITY)

aivancity AI Clinic partner projects (school curriculum; not full-time employment at the partner).

Pangea Summit - AI Clinic 01/2025 - 03/2025 | Paris

- Multimodal RAG (CLIP + CLAP): 9.0/10 reliability, 99% compliance, 75% faster validation; piloted with 10 B2B teams.

Groupe IMAGE et CONCEPT - AI Clinic 09/2024 - 12/2024 | Paris

- Emotion-aware conversational avatar (LLM + CogVideo + Wave2Lip) for naturalistic AI video dialogue.

Atos - AI Clinic 01/2024 - 03/2024 | Paris

- Time-series forecasting (Prophet + XGBoost) for NFT price prediction on blockchain data.

UIZ.CARE - AI Clinic 10/2023 - 12/2023 | Paris

- ICD medical coding system with ~93% accuracy on a WHO-aligned clinical data dictionary.

SELECTED PROJECTS

Emergency Response Twin (live demo and code on GitHub). Cardiac digital twin: live ECG from an ambulance feeds a 1-D CNN that classifies rhythms (e.g. STEMI, VT, VF); a patient twin tracks ST changes and short-term risk; a hospital twin pre-allocates ICU / cath-lab capacity, with a 3D command-center interface.

System Dynamics Model Factory. React + FastAPI + scipy ODE simulator for long-term AI investment scenarios, with optional RAG enrichment.

Chess Piece Detection (YOLOv5). 99.3% mAP50 across 12 piece classes on 1,592 labelled images; real-time inference.

AI Marketing Planner. 4-variant A/B content platform with multilingual output, FAISS knowledge base and live search citation injection.

AI Skill Matcher. PDF resume parser scoring candidates against 500+ job listings, with market-gap dashboards.

TouchlessSHOP. Touch-free e-commerce using MediaPipe hand gestures and Web Speech API for kiosks and hygiene-sensitive environments.

TALKS & COMMUNITY

- Presented **Compar'IA** at the Green-PR Workshop, **ICPR 2026**, Lyon.
- **EuroSciPy 2025**: tutorial (Building Practical Tools for Ethical AI) and talk (Chess Video Analyzer: real-time move detection and narration).
- Presenting at **IEEE DELCON 2026** (BITS Pilani) and **EIRT 2026**.

HONORS

- First in class, aivancity Master's in AI and Data Science: PGE4 (16.39/20, 2024/25) and PGE5 (17.11/20, 2025/26), both with Mention Très Bien.

SKILLS

Programming	Python, SQL, TypeScript, JavaScript, Java, C
LLMs & NLP	PyTorch, Transformers, LangChain, LangGraph, RAG, CLIP, CLAP, Hugging Face, MCP
AI / ML	TensorFlow, Keras, scikit-learn, OpenCV, YOLOv5/v8, RL (PPO/SAC), Prophet, SHAP, LIME, scipy
Systems & MLOps	FastAPI, React, Docker, GCP, Azure, MLflow, DVC, Weights & Biases, FAISS, Pinecone, Neo4j
Languages	English (TOEIC 925/990; Listening C1, Reading B2; research working language)
Certificates	Microsoft Certified: Azure AI Apps and Agents Developer Associate; Microsoft Certified: Azure Data Scientist Associate; AI Product Engineering: From Concept to Market (UC Berkeley SCET)